

GRANT LEYDA

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Portfolio: leydagrant-ux.github.io/PortfolioGrantLeyda

PROFESSIONAL SUMMARY

B.S. Mechanical Engineering graduate from the University of Arkansas (Aerospace Concentration, May 2026) with a 3.37 GPA and hands-on experience in SolidWorks CAD design, FEA structural analysis, and MATLAB. Carried a senior capstone project from concept through manufacturing and on-site delivery: an automated 2,000 lb industrial chassis rotator. SolidWorks CSWA certified. Comfortable talking to customers from two commission sales roles in residential solar, and named to the Chancellor's and Dean's List.

TECHNICAL SKILLS

CAD / Design: SolidWorks (CSWA Certified), Autodesk Inventor, CAD/3D Modeling, 3D Printing

Analysis: FEA (Finite Element Analysis), Von Mises Stress, Displacement and Factor of Safety Analysis, Fluid Flow Simulation (CFD), West Point Bridge Designer

Programming: MATLAB, Computer Methods for Engineering

Manufacturing: Parts Sourcing, Machining, Assembly, BOM Development, Exploded View Drawings

Soft Skills: Team Leadership, Communication, Time Management, Problem Solving, Client Relations

ENGINEERING PROJECTS

Senior Capstone: Cram-A-Lot Manufacturing Chassis Rotator

Fall 2025 to Spring 2026

University of Arkansas, sponsored by JVM / Cram-A-Lot, team design and build project

- Designed and built an automated mechanism to rotate industrial chassis up to 2,000 lbs by 180 degrees, accommodating four different chassis configurations within an 8 ft by 5 ft footprint
- Performed SolidWorks FEA on all critical structural components, including Von Mises stress, displacement, and factor of safety analysis, and verified a minimum factor of safety of 1.6 (1.6 to 3.6 across components) under the 2,000 lb max load at multiple rotation angles
- Led the manufacturing phase, including materials sourcing, machining at UA and JVM facilities, and full system assembly at the sponsor's warehouse, delivering a functional, tested device
- Produced a complete BOM (\$5,432 total cost), exploded view drawings, and full assembly documentation in SolidWorks
- Eliminated reliance on a crane-based rotation process that was causing equipment damage and workflow inefficiency

Full-Stack Session Tracking & Analytics Web App (PWA)

2026

Personal Project | leydagrant-ux.github.io/Bankroll • github.com/leydagrant-ux/Bankroll

- Designed, built, and deployed a multi-user progressive web app (~2,200 lines) with Firebase Authentication and Cloud Firestore, live on GitHub Pages and installable on iOS
- Authored and published Firestore security rules enforcing per-user data isolation and invite-code-only group access, verified end-to-end against the live backend
- Built weekly and all-time leaderboards with an animated SVG visualization scaling continuously to performance, including reduced-motion accessibility support
- Implemented CSV import/export with backward-compatible schema migration so existing user data survives every release

Training Program & Analytics Web App (PWA)

2026

Personal Project | leydagrant-ux.github.io/LockIN • github.com/leydagrant-ux/LockIN

- Built and deployed a multi-user training app on a Cloudflare Workers API with Firebase Authentication and Cloud Firestore, publishing security rules that deny every unauthenticated read and write, verified by probing the live backend directly
- Wrote the progression and analytics engine: double progression with readiness auto-regulation, volume landmarks, estimated 1RM, and Mifflin-St Jeor macro targets, covered by a 369-check regression test suite validated against textbook values
- Integrated the USDA FoodData Central and Open Food Facts APIs with barcode lookup, and modeled a 149-exercise library with equipment-availability filtering and substitution ranking

Additional Coursework Projects

2021 to 2024

- **MATLAB Engineering Computation** (Spring 2024): applied computational methods to solve engineering problems, including numerical analysis and simulation
- **SolidWorks CAD Certification** (Fall 2023): completed a comprehensive CAD design course and earned the SolidWorks CSWA (Certified SolidWorks Associate) credential
- **Bridge Design Project** (Spring 2023): used West Point Bridge Designer to engineer a bridge structure meeting specified load and constraint requirements
- **CAD / Autodesk Inventor** (Fall 2021): completed a design and presentation course with hands-on 3D modeling using Autodesk Inventor

EDUCATION

University of Arkansas, Fayetteville, AR

Graduated May 2026

B.S. Mechanical Engineering | Aerospace Concentration | GPA 3.37

Chancellor's List and Dean's List, Spring 2023 (4.0 GPA)

Member, Sigma Phi Epsilon, Arkansas Alpha Chapter

WORK EXPERIENCE

Olive Garden

Summer 2025

Waiter

- Served guests in a high-volume, fast-paced restaurant environment and coordinated with kitchen and front-of-house staff to keep service running on time

Sunder Energy

May 2024 to January 2025

Commission Sales Associate

- Managed the full sales cycle for residential solar solutions, consistently meeting commission targets in a highly competitive market
- Developed prospecting, objection handling, and closing skills through direct customer engagement

LGCY Power

April 2023 to January 2024

Commission Sales Associate

- Generated solar energy sales through door-to-door outreach and built rapport with customers to educate them on renewable energy products

Lushaj's Pasta & Pizza

June 2021 to Spring 2023

Waiter & Back Prep

- Delivered high-volume customer service in a fast-paced environment and coordinated front and back of house operations

Kroger

March 2020 to October 2020

Customer Service & Cashier

- Provided friendly, efficient service to customers and handled transactions and service issues professionally